CSE7302 - INTERNSHIP

Review-1 Presentation

Development of AI-Driven Robotics: Smart Thrust Bench and Autonomous Navigation Systems

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Under the Supervision of,

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Name of the Program: **COM**
Name of the HoD: **Dr. GopalKrishna Shyam**
Name of the Program Internship Coordinator: **Mr. Shrihari T M**
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About Company or Organization

- **Company Name:** MultiD (Multi Dimensional Technologies (OPC) Pvt. Ltd.)

- **Overview:** MultiD is a specialised engineering and technology firm focusing on **UAVs (Drones)**, **Robotics**, and **AI-driven automation** solutions.

- **Products:** They develop high-performance drone platforms, specialised sensors, and mission-critical software for defence and industrial applications.

- **Clients:** Primary sectors include Defence, Agriculture, and Industrial Automation.

- **Website URL:** https://www.multid.co.in

Working domain or the technology

- **Domain:** Machine Learning & Robotics Integration.

- **Technologies Used:**
  - **Computer Vision:** For people avoidance and obstacle detection.
  - **Sensor Fusion:** Integrating data for the Smart Thrust Bench.
  - **Path Planning:** ML-based algorithms for optimised routing.

- **Importance:** These technologies are critical for increasing the autonomy and safety of unmanned systems in complex environments.

About your team and reporting Manager

- **Team Structure:** Working within the R&D Robotics Division.
- **Role:** Machine Learning and Robotics Intern.
- **Reporting Manager:** Mr. Ganapathi, Senior Robotics Engineer/Lead Developer.
- **Collaboration:** Working alongside hardware engineers for thrust bench calibration and software developers for ML model deployment.

Challenges Faced in Internship

- **Technical Onboarding:** Mastering specific drone telemetry protocols and sensor calibration techniques.

- **Environment Adaptation:** Transitioning from theoretical ML models to real-time hardware constraints.

- **Integration:** Ensuring seamless communication between ML software modules and robotic actuators.

Objectives of the work

- **Smart Thrust Bench:** Develop an automated system to measure and analyze motor/propeller efficiency using real-time data logging.

- **People Avoidance System:** Implement a vision-based ML model to detect and navigate around humans safely.

- **Smart Routing for Farm Fields:** Optimise flight paths for agricultural drones to maximize coverage and battery efficiency.

- **Firing Systems:** Assist in the logic and safety protocols for precision-targeted mechanisms.

Internship Road Map

Review 1	Review 2	Review 3	Review 4	Final Review
Requirement analysis, literature survey, and initial setup of the Smart Thrust Bench.	Development of the People Avoidance ML model and prototype testing for Farm Field routing.	Integration of all modules, final testing on hardware, and performance optimisation.	**System Validation & Edge Deployment:** Fine-tuning ML models for low-latency on hardware, validating thrust bench accuracy against benchmarks, and conducting field trials for farm routing.	**Final Project Handover:** Completion of technical documentation, final demonstration of the firing system safety logic, and performance report on the integrated robotics suite.

Github Link

- URL: https://github.com/MOQA-01/CSE7302---INTERNSHIP

Q&A

Thank you !!